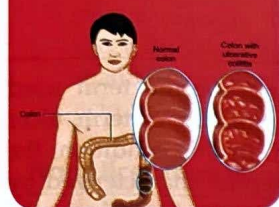


CHAPTER

8



Common Git Diseases

Learning Objectives

- ⇒ To learn about the common occurring gastrointestinal diseases
- ⇒ To know about the common pathophysiology occurring in gastrointestinal diseases
- ⇒ To know the common clinical features in gastrointestinal diseases
- ⇒ To know about ulcerative dyspepsia, non ulcer dyspepsia, irritable bowel syndrome, inflammatory bowel disease in elaborate manner in terms of aetiology, clinical features, method of physical examination and necessary laboratory investigations to diagnose these diseases
- ⇒ To know about the method of physical examination of the patients suffering from above mentioned diseases
- ⇒ To know about the necessary laboratory investigations to be performed to diagnose above diseases

Introduction

Alimentary canal or gastrointestinal tract is a system which starts from mouth to anus with multiple sphincters at different locations to have compartments in the alimentary canal which help in the transformation of unabsorbable macromolecules into the absorbable micromolecules. The gut wall is composed of well defined layers starting from mucosal layer, submucosal layer, muscular layer and serous layer which contribute in specific function of the alimentary canal. As far as the functions of the alimentary canal are concerned, it can be categorized into two major functions – digestion and assimilation of ingested food substances from macromolecules into micromolecules and excretion of undigested food residue from the small intestine to the exterior through anal canal.

As far as the functional anatomy of alimentary canal is concerned, mouth is responsible for the processing of the food in the form of mixing of the food with salivary amylase and sent further to the distal part of the alimentary canal. The food then enters into the oesophagus which further moves into the stomach within seconds. In the stomach, the food is stored for a period of 2 hours where mixing of ingested food with gastric juice containing hydrochloric acid and pepsin, the ingested food is converted into intermediate molecules, thus chyle is formed. After 2 hours, the pyloric antrum opens leading to the movement of chyle into the duodenum. In the

intestine, conversion of macromolecules and intermediate molecules into micromolecules by the intestinal enzymes and bile. The total time duration of the presence of food in the small intestine is upto 6 hours. After the digestion and assimilation of the food substances, the food residue enters into the large intestine. In the large intestine, absorption of water, electrolytes etc occurs and the food residue may remain in the large intestine for a period of 1 to 2 days. Later, the food residue enters the rectum and thence to the anus and later to the exterior when an individual gets defaecation reflex.

As far as overview of gastrointestinal diseases are concerned, it can be subclassified in the following sections.

- ⊛ **Impaired digestion and absorption:** Diseases of stomach, small intestine and large intestine, biliary tree and pancreas may have digestive impairment.
- ⊛ **Altered secretion:** Gastric hypersecretion in Zollinger Ellison syndrome; hyposecretion in case of pernicious anemia.
- ⊛ **Altered gut transit:** This may result mostly because of mechanical obstruction. Esophageal carcinoma, gastric carcinoma, small intestinal obstruction and large intestinal obstruction, achalasia cardia are responsible for the mechanical obstruction.
- ⊛ **Immune dysregulation:** The mucosal inflammation of celiac disease results from dietary ingestion of gluten containing grains. Some patients with food allergy

also exhibit altered immune populations. Eosinophilic esophagitis and eosinophilic gastroenteritis are inflammatory disorders with prominent mucosal eosinophils. Ulcerative colitis and Crohn's disease are manifested due to mucosal injury primarily in the lower alimentary canal. Bacterial, viral and protozoal organisms may produce ileitis or colitis in selected patient populations.

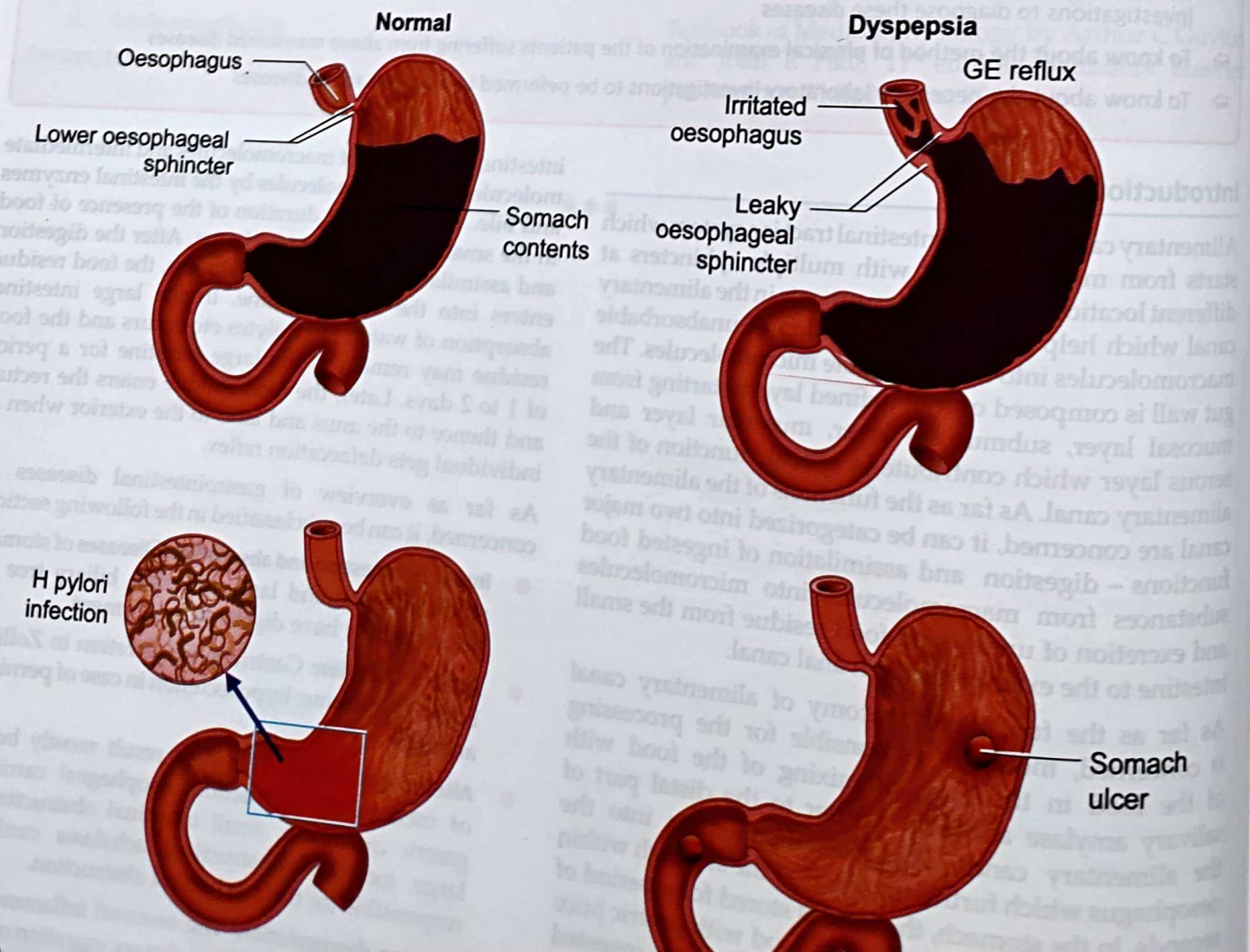
- **Impaired alimentary canal blood flow:** Different organs are at risk for ischemic damage from impaired blood flow. Most commonly observed are intestinal and colonic ischemia, which are consequences of arterial thrombi, emboli, venous thrombi or hypoperfusion from dehydration, sepsis, haemorrhage or reduced cardiac output.
- **Neoplastic degeneration:** All gastrointestinal organs are susceptible for the manifestation of the malignancies. Colon is the most common organ where malignancy can occur.
- **Disorders without obvious structural abnormalities:** Irritable bowel syndrome, non-ulcerative dyspepsia, non cardiac chest pain and functional heartburn

may show altered alimentary canal movement dysfunction.

- **Genetic influences:** some diseases may have an influence of environmental factors, other may have genetic influences. Inflammatory bowel disease (IBD), colonic and oesophageal cancers can be observed in some of the patients as inherited diseases.

Ulcerative Dyspepsia

In western medicine, there is no specific term used as ulcerative dyspepsia (Ref. Figure 13). A term ulcer like functional dyspepsia is used. In this condition, the person will have a symptom complex characterized by epigastric discomfort, burning sensation or pain in the upper abdomen at times associated with eating and often accompanied by belching, nausea or/and vomiting. The difference with non ulcerative dyspepsia, the patient will have abdominal discomfort not related with meal intake. Acid hypersensitivity of the gastric mucosa is considered to be the major causative factor for the manifestation of ulcer like functional dyspepsia.



The prevalence of ulcer like functional dyspepsia ranges from 20-30% and annual incidence of over 1%. It is at least as twice as common from peptic ulcer disease.

- ⊗ Ulcer like functional dyspepsia is common and costly, as it occurs in the patients who have received the treatment to control gastric acidity.
- ⊗ The major differential diagnosis of this disease is acid peptic disease. This condition incorporates gastric ulcer, duodenal ulcer, oesophageal ulcer. This disease manifests in individuals due to imbalance between acid secretion and gastric mucosal defences.

Non Ulcerative Dyspepsia

This is also being termed as functional dyspepsia (Ref. **Figure 14**). In this condition, the structural changes are not noticed. The term dyspepsia refers to persistent or recurrent epigastric pain or meal related upper abdominal pain. This is characterized by early satiety or postprandial fullness. This is different from heartburn (burning sensation in the epigastrium observed till retrosternal region in the chest) is atypical in relation to dyspepsia. This disease is not having any relation to meal association. Endoscopy will be beneficial in the diagnosis of non ulcer dyspepsia.

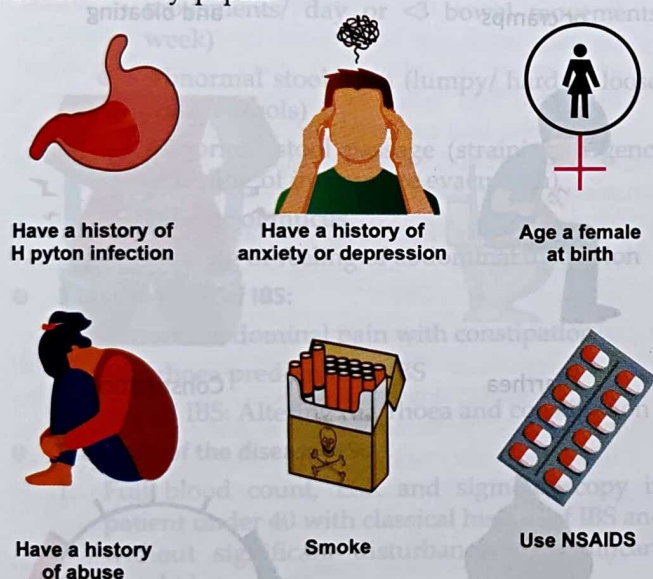


Figure 14 : Non ulcer dyspepsia

- ⊗ Prevalence of the disease is around 25% around the world.
- ⊗ **Aetiological factors:**
 - **Helicobacter pylori infection of the stomach:** This infection is also the probable cause of gastritis, peptic ulcer disease and gastric carcinoma
 - **Disturbed motor function of the stomach:** Delayed gastric emptying, and reduced intestinal motility in the small intestine. This is observed in about 25 to 40% of the patients.

- **Central nervous system disturbances:** Acute stress may lead to nervous disturbances which hampers gastric motility.
- **Diet and environmental factors:** Coffee may induce functional dyspepsia in about 50% of the patients.

⊗ **Clinical features:** The clinical features of functional dyspepsia can be subdivided into those with typical ulcer like symptoms, such as epigastric pain related to meals or waking the patient from sleep (ulcer like dyspepsia) and those with symptoms suggestive of a gut motor disturbance, such as postprandial fullness or early satiety (dysmotility like dyspepsia). Dyspepsia is frequently associated with minor classic reflux symptoms, in particular, heartburn; if reflux symptoms dominate, the diagnosis is symptomatic gastroesophageal reflux disease until proved otherwise.

⊗ **Differential diagnosis:** Differential diagnosis of non ulcer dyspepsia include chronic peptic ulcer disease, gastroesophageal reflux disease with or without oesophagitis and malignancy of the stomach, biliary tract disease, pancreatic disease, drug induced dyspepsia, aerophagia, ischemic heart disease (IHD), intestinal angina, and other disorders such as diabetes mellitus, metabolic disturbances such as hypothyroidism or hypercalcemia.

- **Chronic peptic ulcer disease:** This is associated with mild pain in the epigastric region.
- **Gastroesophageal reflux disease:** This is associated with burning sensation in the epigastrium and retrosternal region.
- **Gastric and oesophageal cancer:** The duration of the illness in these diseases is very shorter i.e. <2 months or less depending upon the nature of the progression of the tumour. The patient may have early satiety and dysphagia and haematemesis, melena.
- **Biliary tract disease:** This is associated with cholelithiasis and obstruction of bile duct with gall stones. In this disease, the person will have pain the right hypochondriac region radiating towards to inferior angle of right scapula. This can be diagnosed by ultrasonography.
- **Pancreatic disease:** In this disease, the person may develop pain in the epigastric region radiating to the back towards the left side. In this disease, the person will have projectile vomiting.
- **Drug induced dyspepsia:** NSAIDs, alendronate, orlistat, iron and potassium supplements, digitalis, theophylline and oral antibiotics, erythromycin and ampicillin are the some of the drugs which leads to the manifestation of dyspepsia.
- **Aerophagia:** Air swallowing with belching

is normal and occurs 3 to 4 times per hour. Excessive, uncontrolled swallowing of air leading to abdominal distension or bloating which improves on occurrence of belching. The diagnosis of aerophagia is confirmed by specific history and can be confirmed by observing excessive air swallowing between meals and repetitive belching.

➤ **Other disorders:**

- ⊗ **Diabetes mellitus:** In this, the person will have postprandial fullness, early satiety, nausea or vomiting, and this can be attributable autonomic dysfunction.
- ⊗ **Hypothyroidism or hypercalcemia:** may induce upper abdominal distress.
- ⊗ **Ischemic heart disease,** of inferior wall of the heart may have upper abdominal distress which is induced by exertion.
- ⊗ **Intestinal angina (chronic mesenteric ischemia):** this can be seen in smokers in particular. In this, the patient will have postprandial fullness and fear of eating and significant weight loss.
- ⊗ **Abdominal wall pain** will have confusion with functional dyspepsia.

Irritable Bowel Syndrome

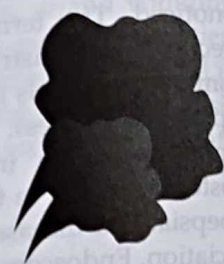
Irritable bowel syndrome (IBS) (Ref. Figure 15) is a functional disease where structural lesions are not observed during the progression of the disease. It consists of group of symptoms related to lower gastro intestinal tract. This is characterised by abdominal discomfort or pain that is accompanied by at least two of the following: relief by defaecation, change in frequency of stools or change in consistency of stool. The cause is unknown, and pathophysiology is poorly understood. Three clinical types are identified depending on the clinical presentation. Diagnosis is clinical.

- ⊗ Irritable bowel syndrome consists of a group of GI symptoms particularly associated with lower bowel in the absence of demonstrable organic pathology.
- ⊗ The etiology of IBS is unknown. No anatomic cause can be found on laboratory tests, x rays or biopsies. Emotional factors, diet, drugs or hormones may precipitate or aggravate GI symptoms. Based on the history, this disease is considered to be purely psychosomatic. Certain physiological factors are known to cause IBS.
- ⊗ **Psychosomatic factors:** Anxiety, depression or somatisation disorder, sleep disturbances, emotional conflict are associated with onset of symptoms and onset of the disease.
- ⊗ **Physiological factors:** A variety of physiological factors

seem to be involved in IBS symptoms. Factors include altered motility, visceral hyperalgesia, various genetic and environmental factors. Constipation may be explained by slower colonic transit and diarrhoea may be explained by faster colonic transit. Visceral hyperalgesia refers to hypersensitivity to normal amounts of intraluminal distension and heightened perception of pain in the presence of normal quantities of intestinal gas. Postprandial abdominal discomfort may be attributed to an exaggerated gastro-colic reflex (the colonic contractile response to a meal). Sometimes, fat ingestion may exaggerate to a meal). Rectal sensitivity is increased during menstruation but not during other phases of menstrual cycle.



Abdominal pain or cramps



Excess gas and bloating



Diarrhea



Constipation



Mucus in your poop (may look whitish)



Feeling like you're unable to empty your bowels after pooping

Figure 15 : Clinical features of irritable bowel syndrome

- ⊗ IBS tends to begin in teens and 20s, causing bouts of symptoms that recur at irregular periods. Onset in late adult life is less common, but not rare. Symptoms may rarely rouse the sleeping patient. symptoms are often triggered by food, particularly fat or by stress. Patients will have abdominal discomfort in

the lower quadrant, steady and cramping in nature, and relieved by defaecation. Abdominal discomfort is temporarily associated with alteration in stool frequency. Patients may have symptoms of abnormal stool passage (straining, urgency or feeling of incomplete evacuation), pass mucus or complain of bloating or abdominal distension. Many patients may also have symptoms of dyspepsia. Extraintestinal symptoms (fatigue, fibromyalgia, sleep disturbances, chronic headaches) are common.

⊗ **Diagnosis is clinical**, based on Rome criteria.

⊗ **Rome criteria:** Recurrent abdominal pain or discomfort for at least 3 days per month in last 3 months (with symptoms onset at least 6 months prior to diagnosis) associated with two or more of the following:

1. Improvement after defaecation
2. Onset associated with change in frequency of stools
3. Onset associated with change in form (appearance) of stools
4. Other symptoms that are not essential but support the diagnosis of IBS
 - ⊗ Abnormal stool with frequency (>3 bowel movements/ day or <3 bowel movements/ week)
 - ⊗ Abnormal stool form (lumpy/ hard or loose/ watery stools)
 - ⊗ Abnormal stool passage (straining, urgency or feeling of incomplete evacuation)
 - ⊗ Passage of mucus
 - ⊗ Bloating or feeling of abdominal distension

⊗ **3 clinical types of IBS:**

1. Chronic abdominal pain with constipation
2. Diarrhoea predominant IBS
3. Mixed IBS: Altering diarrhoea and constipation

⊗ **Diagnosis of the disease IBS:**

1. Full blood count, ESR and sigmoidoscopy in patient under 40 with classical history of IBS and without significant disturbance or significant weight loss.
2. Rectal biopsy in patients with diarrhoea. Changes of inflammatory bowel disease or less common types of colitis may be found in a normal rectum.
3. In patients with chronic diarrhoea, further investigations to rule out coeliac disease, Crohn's disease.
4. Rome Criteria of IBS is followed.

Differential diagnosis of IBS: Lactose intolerance, drug induced diarrhoea, post-cholecystectomy diarrhoea, laxative abuse, parasitic disease, eosinophilic gastritis or enteritis, microscopic colitis and early inflammatory bowel disease.

Inflammatory Bowel Disease

The term 'inflammatory bowel disease' or IBD is commonly used to include 2 idiopathic bowel diseases having many similarities, but have distinctive morphological appearance. These diseases are Crohn's disease (regional enteritis) and ulcerative colitis.

1. Crohn's disease is also termed as regional enteritis is an idiopathic chronic ulcerative IBD. This is characterised by transmural, non-caseating granulomatous inflammation, affecting most commonly the segment of terminal ileum or colon, though any part of the alimentary canal may be involved.
2. Ulcerative colitis is an idiopathic form of acute and chronic ulcero-inflammatory colitis. This disease affects mucosa and submucosa of the rectum and descending colon. Sometimes, the whole large intestine may be afflicted.

Etiopathogenesis: The exact etiology of the IBD is not known. Genetic factors such as occurrence of the disease 3 to 20 times higher in first degree relatives, 50% of occurrence of the disease in monozygotic twins, defective regulation of immune suppression or activation of CD4+ cells or poor evidence of influence of Mycobacterium paratuberculosis or serious mental stress are often considered as triggering factors for the worsening of the symptoms in the patients of IBD. This indicates the multifactorial occurrence of the disease.

Crohn's disease:

- ⊗ This disease may involve any portion of the alimentary canal, but it commonly affects 15-25 cms of terminal ileum which may into the caecum and sometimes into the ascending colon.
- ⊗ This is often presented with multiple, well demarcated segmental bowel involvement with intervening uninvolved skip areas. The wall of the affected bowel segment is thick and hard, resembling a hose pipe. Serosa may be studded with ulcers, granulomas.
- ⊗ Histologically, the tissue of the affected part contains chronic inflammatory cells (lymphocytes, plasma cells and macrophages) is the classical microscopic feature.
- ⊗ The most common initial manifestation is chronic diarrhoea with abdominal pain, fever, anorexia and weight loss. The abdomen is tender, and a mass or fullness may be palpable. Gross rectal bleeding is unusual except in isolated colonic disease. With recurrent disease, symptoms may vary. Pain is most common and occurs with both simple recurrence and abscess formation. In progression of the disease, there will be formation of stenosis of the colon. Stenotic segments of colon may lead to intestinal obstruction associated with colicky pain, distension of abdomen, obstipation and vomiting. Adhesions can previous surgery can also lead to intestinal obstruction.

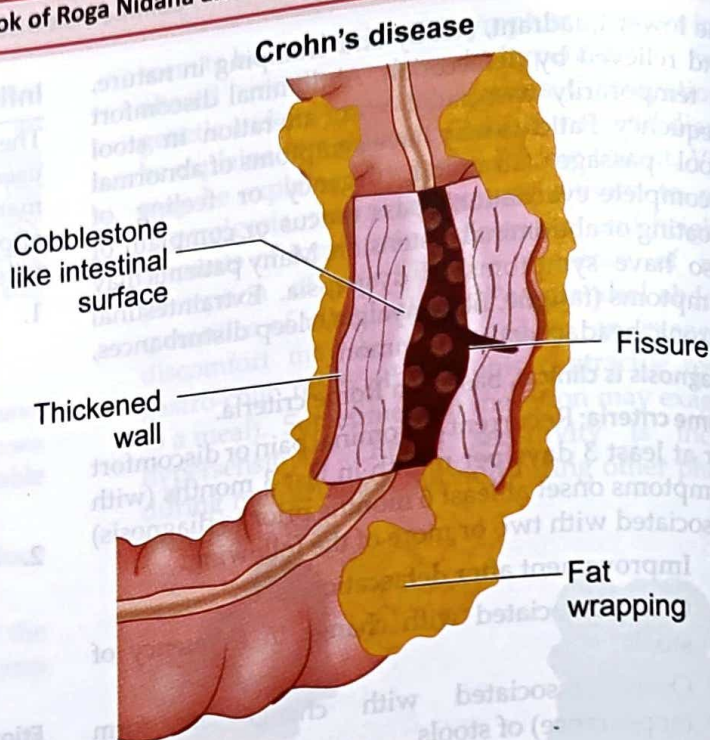
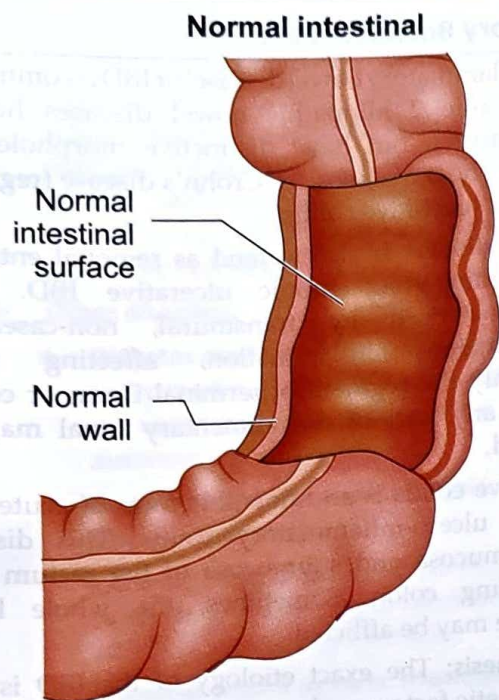


Figure 16 : Normal large intestine and intestine of Crohn's disease

Chronic Crohn's disease causes a variety of systemic symptoms including fever, weight loss, undernutrition and extraintestinal manifestations.

Crohn's disease is classified into 3 principal patterns

1. **Inflammatory pattern:** which after several years can turn into stenotic or obstructing and penetrating or fistulising

2. Stenotic or obstructing lesion which lead to intestinal obstruction
3. Penetrating or fistulising lesion which lead to fistula formation in the colon

Complications of Crohn's disease are

1. Malabsorption due to impaired absorption of fat, vitamin B12, proteins and electrolytes
2. Fistula formation may occur in long standing cases
3. Stricture formation due to extensive fibrosis in the affected bowel wall.
4. Development of malignancy in the small intestine.

Diagnosis of Crohn's disease:

1. Barium X ray of stomach, small intestine, colon
2. Abdominal CT scan
3. Upper endoscopy and colonoscopy

Ulcerative colitis:

Classically, this disease begins in the rectum, and it continues upwards to sigmoid colon, descending colon, transverse colon and sometimes, whole colon

can be involved.

In this disease, the affliction is usually continuous, no skip areas are seen as in Crohn's disease. The appearance of colon may vary depending upon the stage and intensity of the disease because of remissions and exacerbations. Mucosa shows linear and superficial ulcers and not penetrating the muscular layer of the colon.

Histologically, the tissue of the affected part is characterised by alternating active disease process and resolving colitis.

Symptoms and signs: Bloody diarrhoea of varied intensity and duration is interspersed with asymptomatic intervals. Usually an attack begins insidiously, with increased urgency to defaecate, mild lower abdominal cramps and blood and mucus in the stools. Some cases start after the exposure to infection such as amoebiasis or bacillary dysentery.

When ulceration is confined to sigmoid colon and rectum, the stool may be normal or hard and dry, but rectal discharges of mucus is loaded with RBCs and WBCs accompany or occur between bowel movements. If ulceration extends proximally, stools become looser and the patient may have > 10 bowel movements per day, often with severe cramps and distressing rectal tenesmus, without respite at night. The stools may be watery or contain mucus and frequently consist of blood and mucus.

Toxic or fulminant colitis manifests initially with sudden violent diarrhoea, high grade fever upto 40° C (104° F), abdominal pain, signs

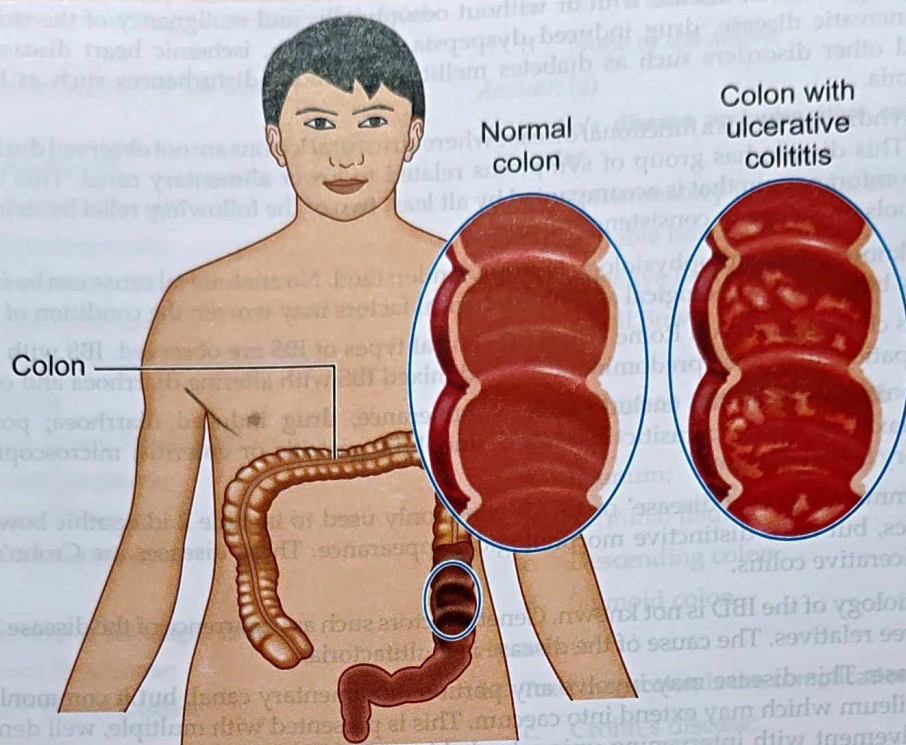


Figure 17 : Normal colon and colon of ulcerative colitis

of peritonitis such as rebound tenderness and profound toxemia.

- Systemic symptoms which occur in extensive ulcerative colitis include malaise, fever, anemia, anorexia and weight loss. Extraintestinal manifestations such as joint and skin complications in particular are most common when systemic symptoms are present.

❖ **Complications of ulcerative colitis:**

- **Toxic megacolon (Fulminant colitis):** This is acute fulminating colitis in which the affected colon

is thin walled and dilated and is prone to perforation and fecal peritonitis.

- Carcinoma may develop in long standing cases of ulcerative colitis of more than 10 years duration.
- Perianal fistula and stricture formation in the affected part is rarely seen.

❖ **Diagnosis of ulcerative colitis:**

- Stool cultures and microscopy examination of stool
- Sigmoidoscopy with biopsy

Key summary of the chapter

- ❖ This chapter discusses about the brief outline of description of normal and abnormal functioning of alimentary canal. This is one system which commences from mouth to anus, 9 meters long, helps in ingestion, storage, digestion, assimilation and excretion of the food substances. Due to various aetiological factors, the person may have abnormalities in above said functions of alimentary canal.
- ❖ The pathological events in the alimentary canal can be categorised into impaired digestion and absorption; altered secretion; altered gut transit; immune dysregulation; impaired alimentary canal blood flow; neoplastic degeneration; disorders without structural abnormalities and genetic factors induced.
- ❖ Ulcerative dyspepsia is a ulcer like functional dyspepsia, will have a symptom complex characterised by epigastric discomfort, burning sensation or pain in the upper abdomen. This is usually associated with eating.
- ❖ **Nonulcerative dyspepsia:** This is also termed as functional dyspepsia. No structural changes are noticed in this disease. This is characterised by early satiety or postprandial fullness. This is not having any relation to meal association. H pylori infection, disturbed motor function of the stomach, central nervous system disturbances such as mental stress, coffee may induce nonulcerative dyspepsia.
- **Differential diagnosis:** Differential diagnosis of non ulcer dyspepsia include chronic peptic ulcer disease,

gastrooesophageal reflux disease with or without oesophagitis and malignancy of the stomach, biliary tract disease, pancreatic disease, drug induced dyspepsia, aerophagia, ischemic heart disease (IHD), intestinal angina, and other disorders such as diabetes mellitus, metabolic disturbances such as hypothyroidism or hypercalcemia.

- Irritable bowel syndrome (IBS) is a functional disease where structural lesions are not observed during the progression of the disease. This disease has group of symptoms related to lower alimentary canal. This is characterised by abdominal discomfort or pain that is accompanied by at least two of the following: relief by defaecation, change in frequency of stools or change in consistency of stool.
 - Cause is unknown, and pathophysiology is poorly understood. No anatomical cause can be found on laboratory tests, x rays, biopsies. Psychological and physiological factors may worsen the condition of IBS.
 - Diagnosis is clinical, based on Rome criteria. 3 clinical types of IBS are observed: IBS with chronic abdominal with constipation; diarrhoea predominant IBS and mixed IBS with altering diarrhoea and constipation.
 - Differential diagnosis of IBS include lactose intolerance; drug induced diarrhoea; post-cholecystectomy diarrhoea; laxative abuse; parasitic disease; oesinophilic gastritis or enteritis; microscopic colitis and early inflammatory bowel disease.
- The term 'inflammatory bowel disease' or IBD is commonly used to include 2 idiopathic bowel diseases having many similarities, but have distinctive morphological appearance. These diseases are Crohn's disease (regional enteritis) and ulcerative colitis.
 - The exact etiology of the IBD is not known. Genetic factors such as occurrence of the disease 3 to 20 times higher in first degree relatives. The cause of the disease is multifactorial.
 - **Crohn's disease:** This disease may involve any portion of alimentary canal, but it commonly affects 15-25 cms of terminal ileum which may extend into caecum. This is presented with multiple, well demarcated segmental bowel involvement with intervening uninvolved skip areas. The wall of the affected bowel segment is thick and hard, resembling a horse pipe. Serosa may be studded with ulcers, granulomas. The most common initial manifestation is chronic diarrhoea with abdominal pain, fever, anorexia and weight loss. Complications of the Crohn's disease are malabsorption syndrome, fistula formation, stricture formation and malignancy of small intestine. Diagnosis of the disease is done by barium X ray of stomach, small intestine, colon; abdominal scan and upper endoscopy and colonoscopy.
 - **Ulcerative colitis:** Classically, this disease begins in the rectum, and it continues upwards to sigmoid colon, descending colon, transverse colon and sometimes whole colon can be afflicted. In this disease, the affliction is continuous, no skip areas seen like Crohn's disease.
 - The appearance of colon may vary depending upon the stage and intensity of the disease because of remissions and exacerbations. Symptoms and signs include blood mixed diarrhoea of varied intensity and duration is interspersed with asymptomatic intervals. Usually, an attack begins insidiously, with increased urgency to defaecate, mild lower abdominal cramps and blood and mucus in the stools. Fever, malaise, anaemia, anorexia and weight loss; extra intestinal manifestations such as joint and skin complications are also observed. Complications of the disease are toxic megacolon; carcinoma of rectum or colon; perianal fistula and stricture. Diagnosis of the disease is done by stool culture; microscopy examination of stool and sigmoidoscopy with biopsy.

Self-Assessment: Questions for Review

Short essay questions

- Explain irritable bowel syndrome
- Describe the pathology of irritable bowel syndrome
- Explain the causative factors of irritable bowel syndrome
- Explain in brief ulcerative dyspepsia
- Describe nonulcerative dyspepsia
- Explain Crohn's disease
- Describe ulcerative colitis
- Explain the differences between Crohn's disease and ulcerative colitis

MCQs

1. The chyme enters into small intestine, which reaches caecum in
 - a. 10 minutes;
 - b. 6 hours;
 - c. 10 hours;
 - d. 1 - 2 days

Answer: (b)

2. Which of the following is false in case of functional dyspepsia?
 - a. Also termed as nonulcerative dyspepsia;
 - b. The symptoms are not having any association with food intake;

- c. The symptoms are having association with food intake;
- d. All of the above statement are true

Answer: (c)

3. Crohn's disease is a type of

- a. Ulcerative dyspepsia;
- b. Nonulcerative dyspepsia;
- c. Inflammatory bowel disease;
- d. Irritable bowel syndrome

Answer: (c)

4. The term IBS refers to

- a. Inflammatory Bowel Syndrome;
- b. Irritable Bowel Syndrome;
- c. Infective Bowel Disease;
- d. Ischemic Bowel Syndrome

Answer: (b)

5. Rome criteria will help in the diagnosis of

- a. Irritable Bowel Syndrome;
- b. Inflammatory Bowel Disease;
- c. Ulcerative colitis;
- d. Crohn's disease

Answer: (b)

6. Inflammatory Bowel Disease is a disease of

- a. Stomach;
- b. Small intestine;
- c. Large intestine;
- d. None of the above

Answer: (c)

7. As per Rome criteria, the duration of abdominal pain should persist for

- a. 3 days to 3 months;
- b. 7 days to 2 months;
- c. 15 to 20 days;
- d. 1 to 6 months

Answer: (a)

8. Irritable Bowel Syndrome begins at the age of

- a. 30 to 40 years;
- b. 15 to 20 years;
- c. 60 to 70 years;
- d. 80 to 90 years

Answer: (b)

9. Which of the following is having patchy appearance in pathological changes?

- a. Crohn's disease;
- b. Ulcerative colitis;

- c. Both a and b;
- d. None of the above

Answer: (a)

10. Crohn's disease and ulcerative colitis are categorised under

- a. Inflammatory Bowel Disease;
- b. Irritable Bowel Syndrome;
- c. Infective Bowel Disease;
- d. Small Bowel Disease

Answer: (a)

11. Most common part of the intestine involved in Crohn's disease is

- a. Jejunum;
- b. Terminal ileum;
- c. Descending colon;
- d. Sigmoid colon

Answer: (b)

12. Toxic megacolon is the complication of

- a. Crohn's disease;
- b. Irritable Bowel Syndrome;
- c. Ulcerative colitis;
- d. Tropical sprue

Answer: (c)

13. Malabsorption syndrome is caused by

- a. Crohn's disease;
- b. Ulcerative colitis;
- c. Both a and b;
- d. None of the above

Answer: (a)

14. Ulcerative colitis begins from which part of the large intestine?

- a. Rectum;
- b. Sigmoid colon;
- c. Descending colon;
- d. Transverse colon

Answer: (a)

15. Aerophagia is one of the differential diagnosis for the diagnosis of

- a. Functional dyspepsia;
- b. Ulcerative dyspepsia;
- c. Inflammatory bowel disease;
- d. Irritable bowel syndrome

Answer: (a)

Chapter resources

- ❖ <https://patient.gastro.org/dyspepsia/>
- ❖ <https://thegidocs.com/peptic-ulcer-and-non-ulcer-dyspepsia/>
- ❖ <https://my.clevelandclinic.org/health/diseases/4342-irritable-bowel-syndrome-ibs>
- ❖ <https://www.drugwatch.com/health/digestive-health/crohns-disease/>
- ❖ <https://mydr.com.au/gastrointestinal-health/ulcerative-colitis/>
- ❖ Golwalla's medicine for students by Aspi F Golwalla and Sharukh A Golwalla, edited by Milind Y Nadkar, 25th edition, Jaypee Brothers Medical Publishers (P) Ltd, New Delhi, 2017
- ❖ The Merck Manual of Diagnosis and Therapy by Robert S Porter and Justin L Kaplan, 19th edition, Merck Sharp & Dohme Corp, New Jersey, 2011
- ❖ Chamberlain's symptoms and signs in clinical medicine by Colin Ogilvie and Christopher C Evans, 12th edition, Butterworth-Heinemann Publishers, 1997
- ❖ Best & Taylor's physiological basis of medical practice by O P Tandon and Y Tripathi, 13th edition, Wolters Kluwer Health (India) Pvt. Ltd, Gurgaon, 2012
- ❖ Textbook of Medical physiology by Arthur C Guyton and John E Hall, 11th edition, Saunders Elsevier publications, Philadelphia, 2006
- ❖ Cecil's medicine edited by Lee Goldman and Dennis Ausiello, 23rd edition, Saunders Elsevier publications, Philadelphia, 2007
- ❖ <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4531982/>
